

Concentration determination of a molecule using UV-Vis spectrophotometer

Experiment #4



Why Cherries are red and blueberries are blue ?

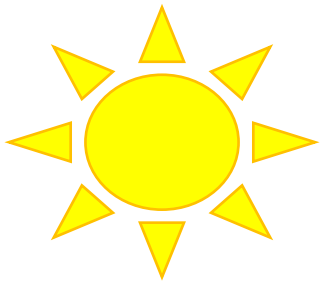
Colour

The color of an object we see is due to the wavelengths that reach our eyes reflected, transmitted, emitted or scattered.

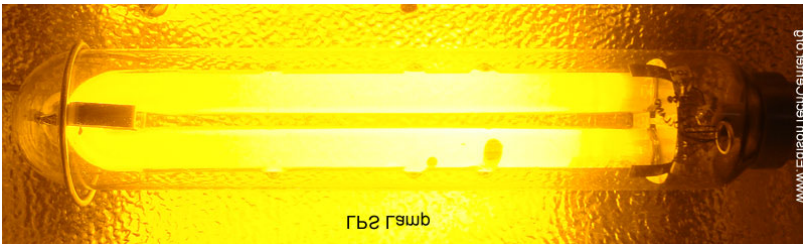
Examples

- Emission

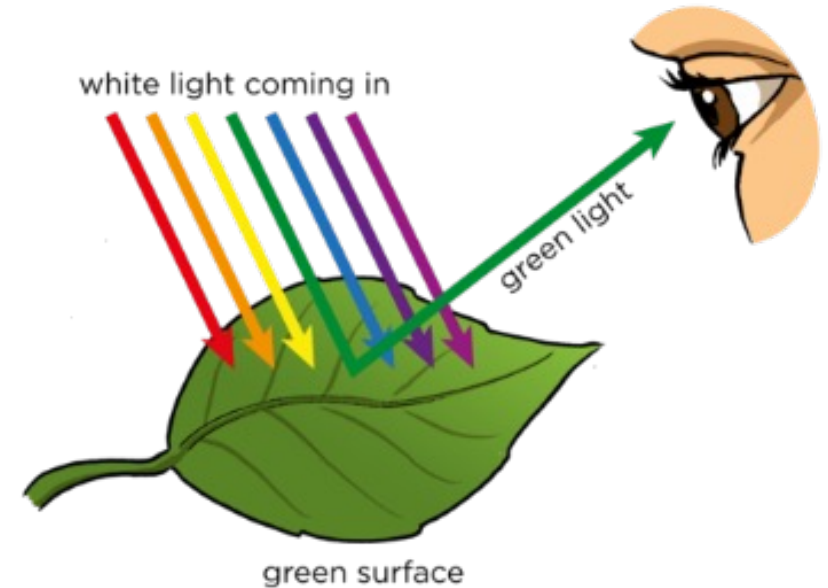
Continuous Spectra



Discrete Spectra



- Reflection



<http://www.mstworkbooks.co.za/natural-sciences/gr8/images/gr8ec04-gd-0027.png>

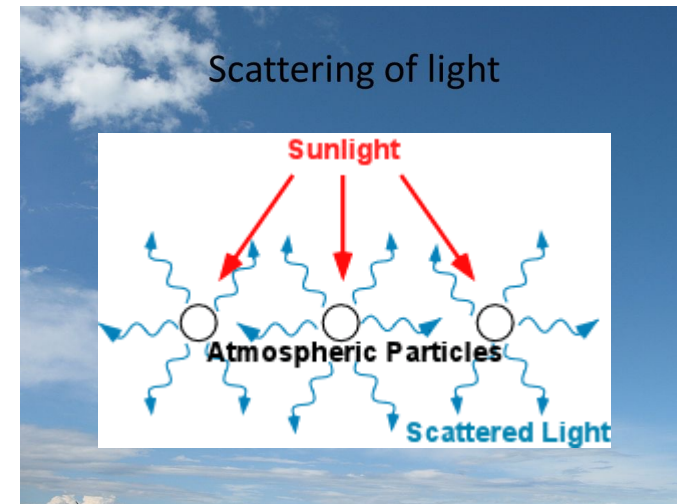
Cont.....

- Absorptions and Transmission



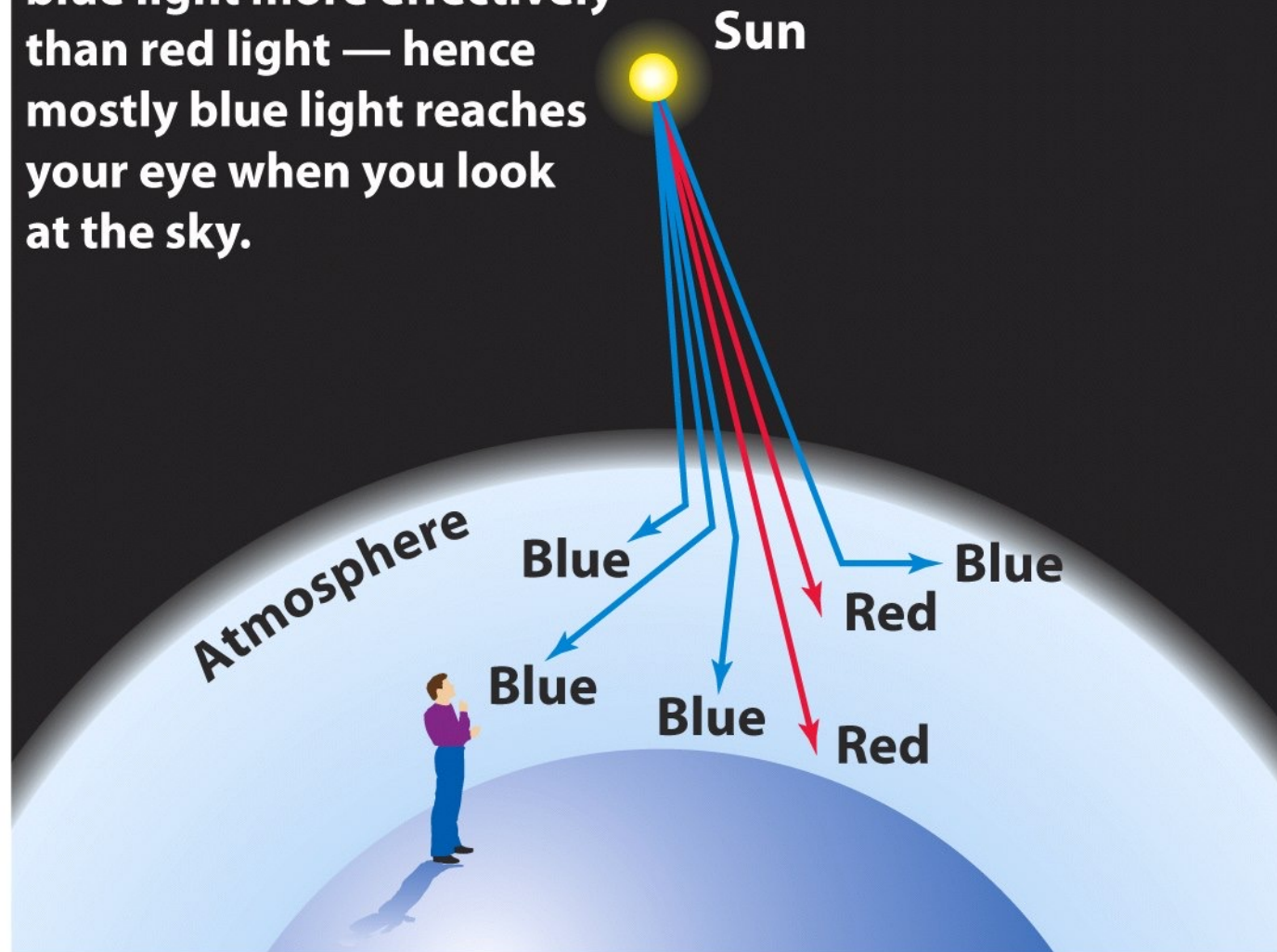
http://3.bp.blogspot.com/-lcOuWVybErE/UPX_8nLRQ9I/AAAAAAAAAMOW/89_uBFv3zmE/s1600/absorption.jpg

- Scattering

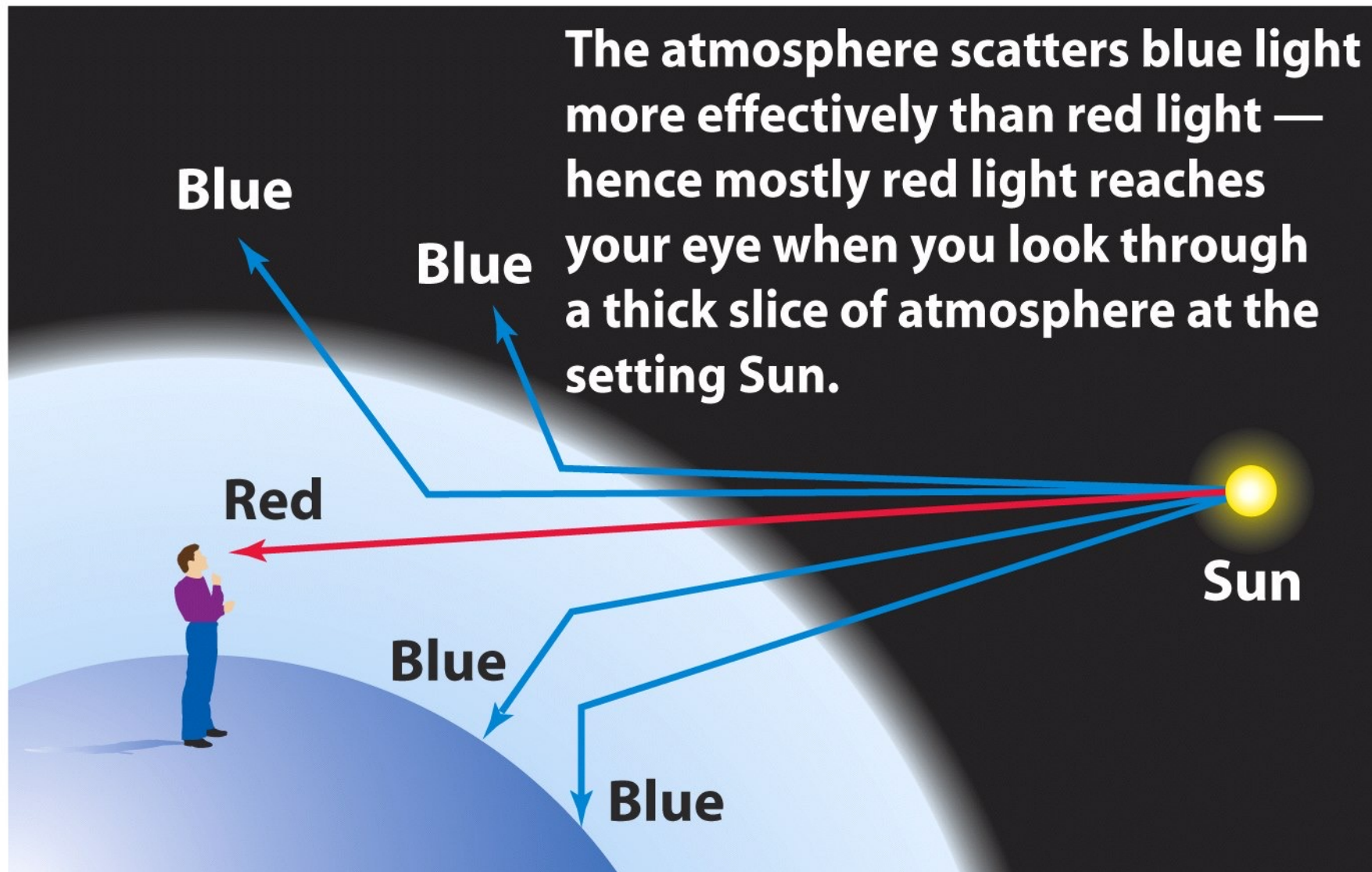


<http://ww2010.atmos.uiuc.edu/guides/mtr/opt/mch/gifs/sct1.gif>

**The atmosphere scatters
blue light more effectively
than red light — hence
mostly blue light reaches
your eye when you look
at the sky.**



(a) Why the sky looks blue

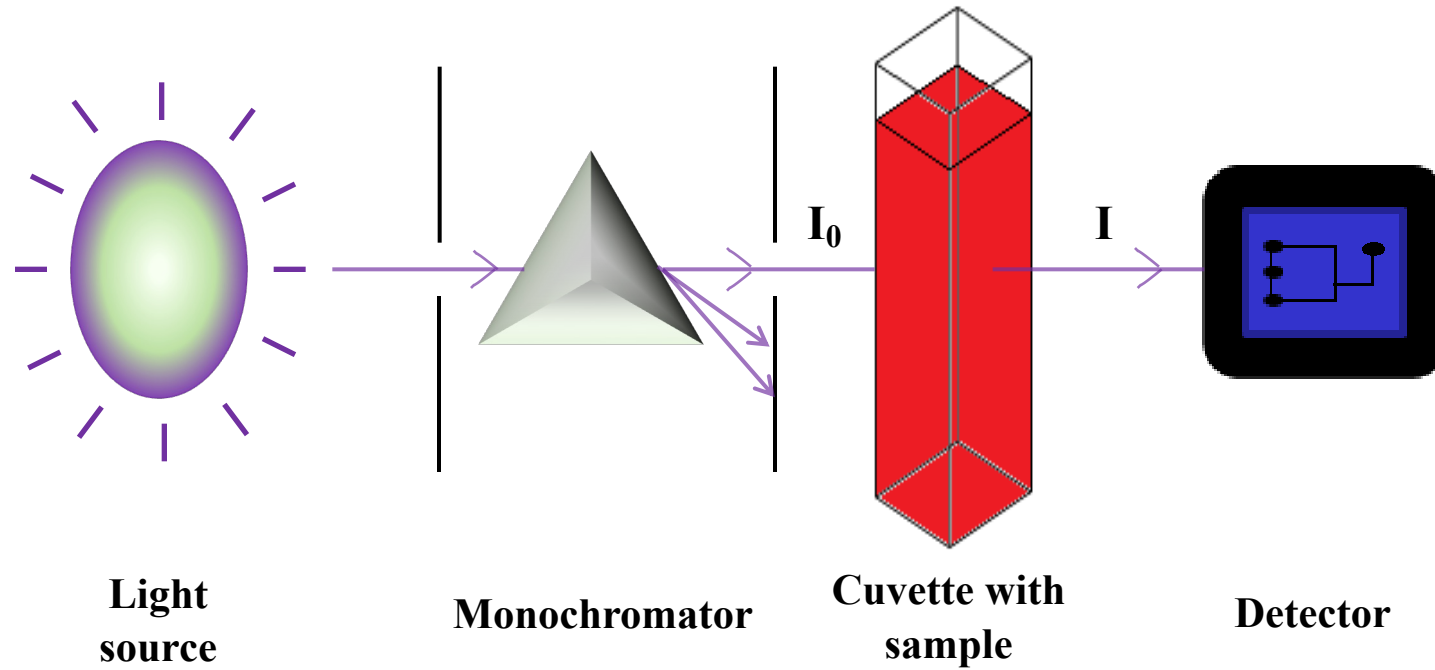


(b) Why the setting Sun looks red

Spectrum

Instead of eye, wavelengths/frequencies absorbed, emitted, transmitted or scattered by an object/atom/molecule can also be observed by placing a detector in appropriate direction.

Absorption Experimental Set Up



UV-Visible spectroscopy

- Absorption of light in the UV/Visible part of the spectrum (210 – 900 nm).
- The transitions that result in the absorption of electromagnetic radiation in this region of the spectrum are transitions between electronic energy levels.
- Generally, the most probable transition is from highest occupied molecular orbital (HOMO) to lowest occupied molecular orbital (LUMO).

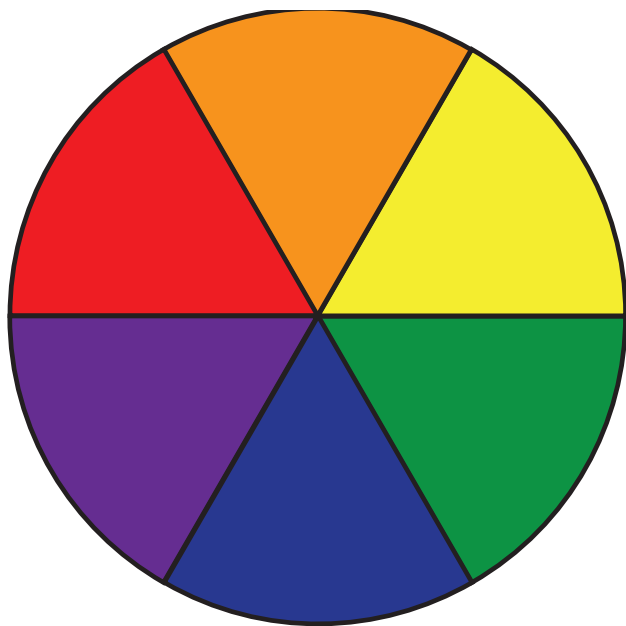


Figure 1. Color wheel

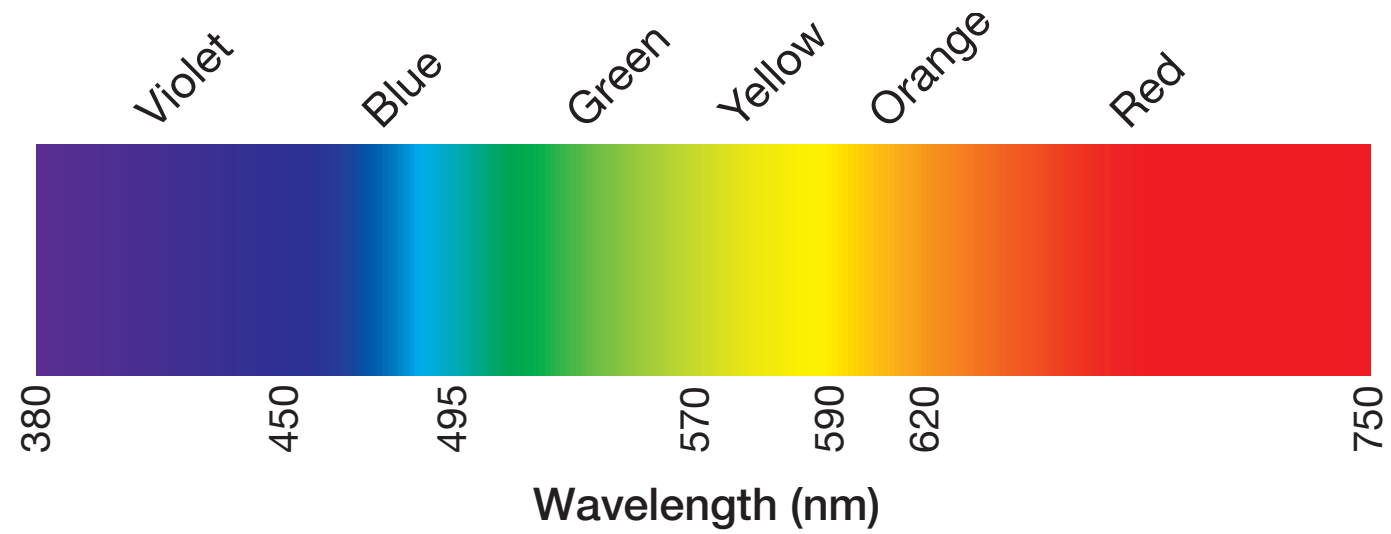
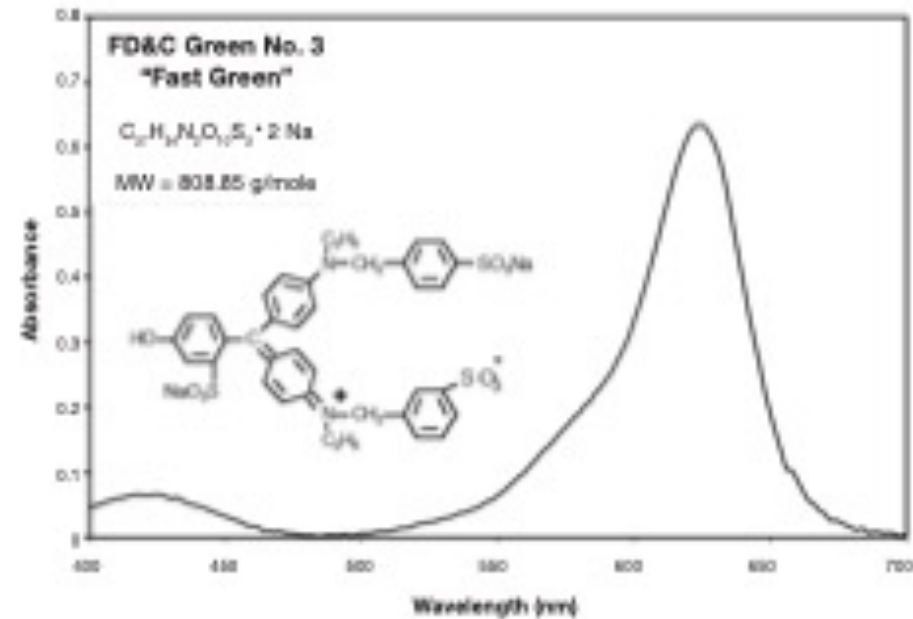


Figure 2. Visible spectrum

- You will identify a peak or peaks in the scan and record the wavelength of each peak. Officially, the wavelength at the top of the peak is called the “wavelength of maximum absorbance”, which is abbreviated to λ_{max} (spoken as “lambda max”).



IN DETECTION OF METALS

Metal	Method	λ_{max} (nm)
Aluminium	React with Eriochrome cyanide R dye at pH 6 to form red to pink complex	535
Arsenic	On reaction with Zn, arsenic reduces to AsH ₃ which react with silver to form red color complex diethyldithiocarbamate	535
Cadmium	On reaction with chloroform solution of dithiozone in presence of NaOH give pink to red complex	518
Chromium	Cr(VI) react with chloroform solution pf dithiozide to give red-violet complex	540
Lead	Dithiozone in presence of with NH ₃ give cherry red complex	510
Mercury	On reaction with chloroform solution of dithiozone under acidic conditions give orange complex	492
Copper	React with neocuprine in neutral to slightly acid solution and extraxt into CHCl ₃ /CH ₃ OH; forms yellow complex	457

ANALYSIS OF NARCOTICS AND DRUG TESTING

Many illegal drugs contain aromatic groups that produce characteristic UV/Vis spectra.

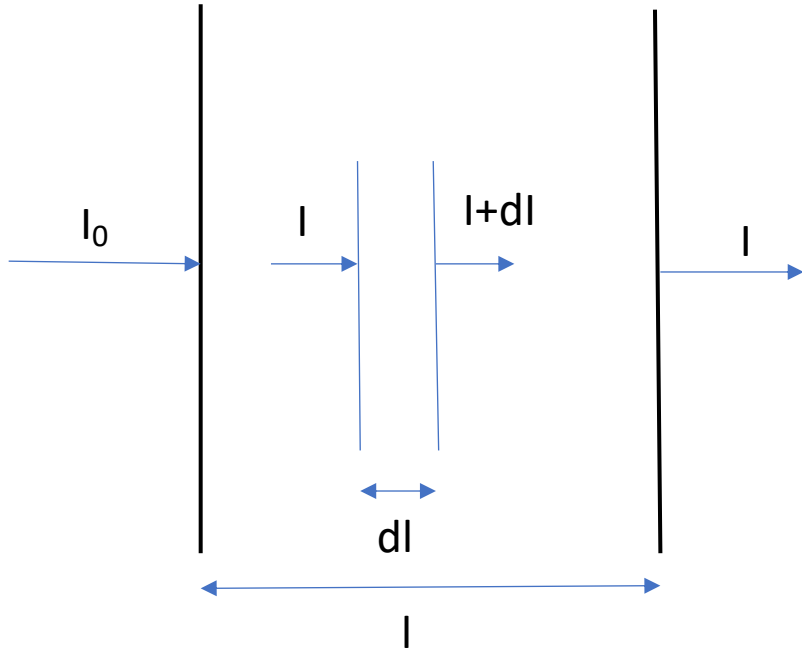
S.No.	Drug	λ_{max} (nm)
1	Cocaine in 0.2 N H ₂ SO ₄	233
2	Codenine in water	284
3	Morphine in acidic solution	285
4	Nicotine in 0.2 N H ₂ SO ₄	260
5	Amphetamine in methanol	258
6	Ectasy or MDMA in ethanol	286
7	Caffeine in 0.2 N H ₂ SO ₄	271

Part 1. Scan the dyes

Data Table 1

Color of Solution	λ_{max} (nm)	Absorbance
Red		
Yellow		
Green		
Blue		

Lambert's Beer Law



$$\frac{dI}{I} = -kcdl$$

$$\int_{I_0}^I \frac{dI}{I} = \int_0^l -kcdl$$

$$\ln \frac{I_0}{I} = kcl$$

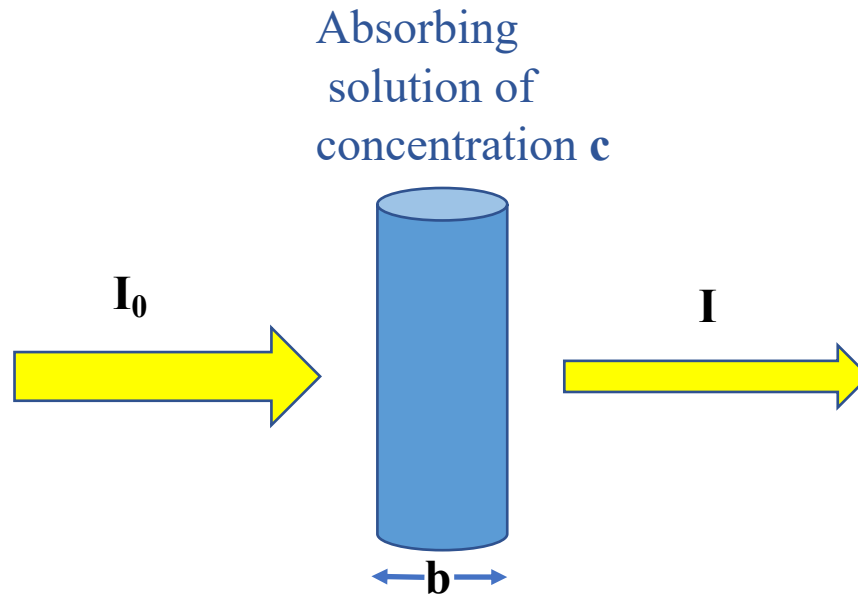
$$\log \frac{I_0}{I} = 2.303kcl = \epsilon cl$$

Lambert Beer Law

Lambert-Beer law states that the quantity of light absorbed by a substance dissolved in a fully transmitting solvent is directly proportional to the **concentration** of the substance and the **path length** of the light through the solution.

$$I = I_0 10^{-\epsilon cb}$$

I_0 and I are incident and transmitted intensities



$$A = \log (I_0/I)$$

A is absorbance

$$T = I/I_0$$

T is transmittance

Absorbance and the Beer – Lambert Law

The intensity of light decreases exponentially as it passes through an absorbing medium. The **absorption**, A is defined as $\log(I_0/I)$, where I_0 is the intensity of the incident light and I is the intensity of the transmitted light.

The Beer – Lambert law quantifies the process of absorption. At a given wavelength,

$$A = \epsilon cl$$

ϵ is the **Molar Extinction Co-efficient**. It is a constant for the absorbing species and defines the absorption of the species at a particular wavelength. It is determined by the number & type of chromophores present in each molecule of the absorbing species.

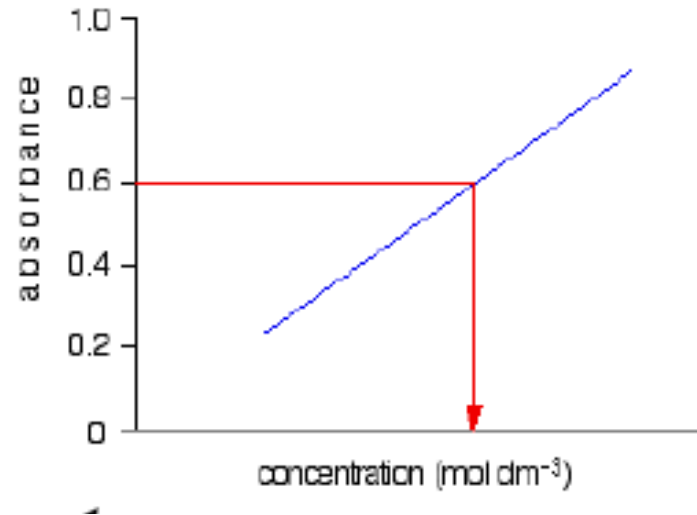
c is the **Concentration** of the sample in the cuvette.

l is the **length of the light path** through the sample.

Absorbance is an additive property. The absorbance of a mixture of n species is given by,

$$A_{mixture} = \sum_{i=0}^n A_i$$

According to the Beer-Lambert Law, absorbance is proportional to concentration, and Absorbance versus concentration plot is a straight line.



Quantitative Analysis

- UV absorption spectroscopy can be used for the quantitative determination of compounds that absorb UV radiation. This determination is based on Beer's law which is as follows.

$$A = \log I_0 / I_t = \log 1/T = -\log T$$

$$A = \epsilon cl$$

- Where ϵ is extinction co-efficient, c is concentration, and l is the length of the cell that is used in UV spectrophotometer.
- Example: for estimation of aspirin (ASP), impurities from such as salicylic acid (SAL) and heavy metal ions (HMI) in ASP tablets

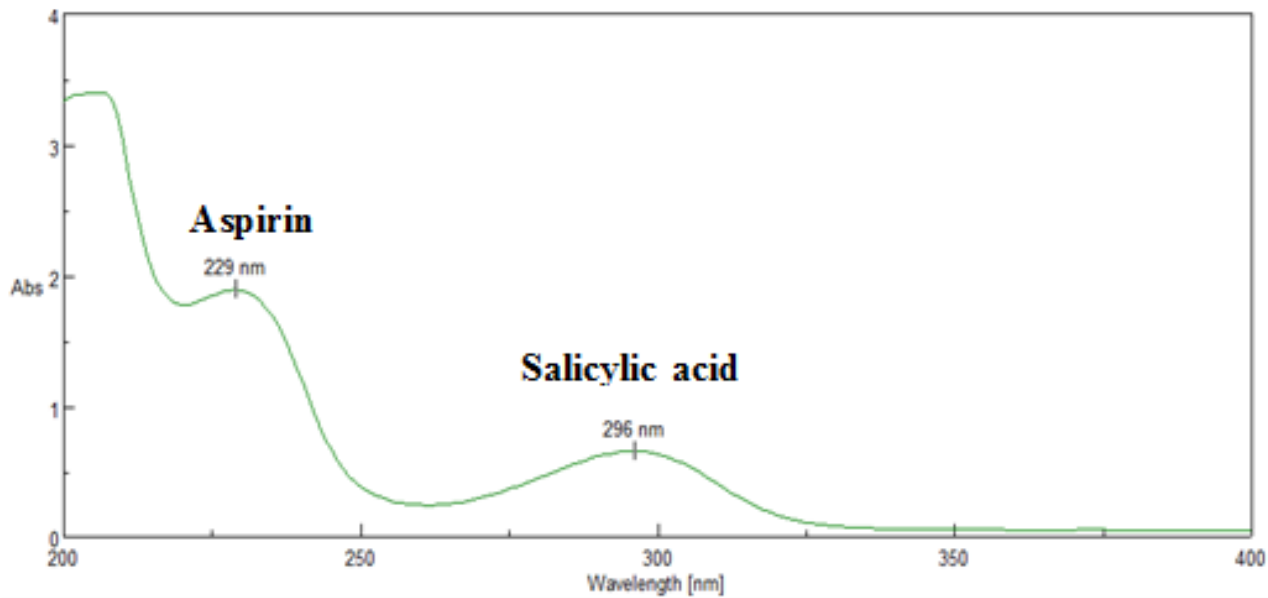
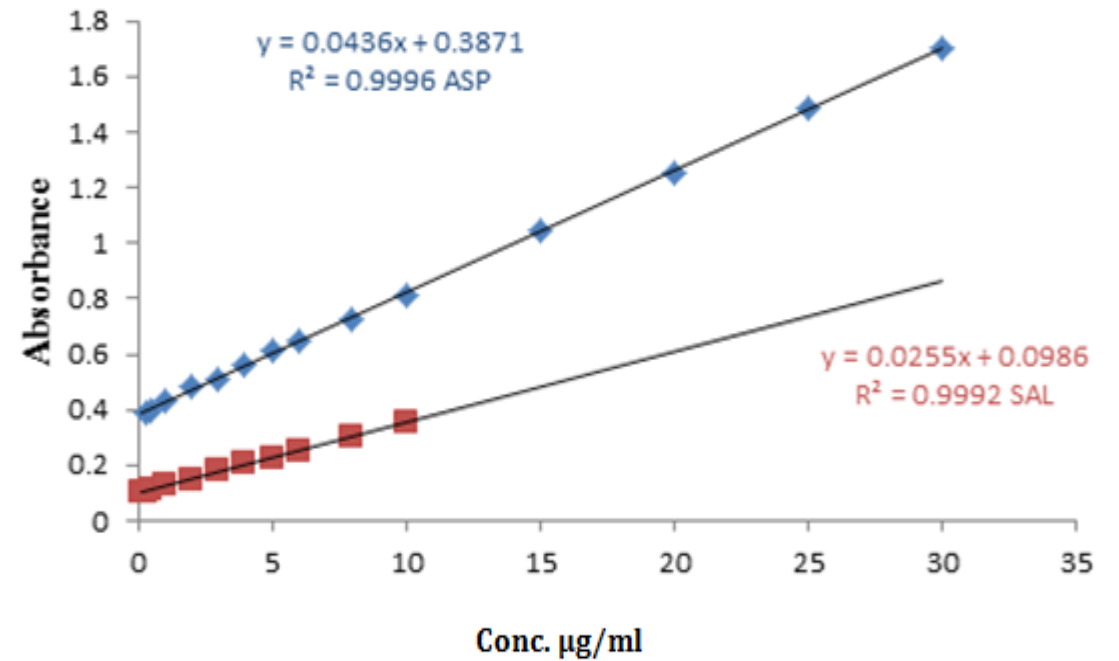


Fig.3 : UV-Vis spectrum of aspirin and salicylic acid mixture.

<https://innovareacademics.in/journals/index.php/ijap/article/download/26820/15657/137236>

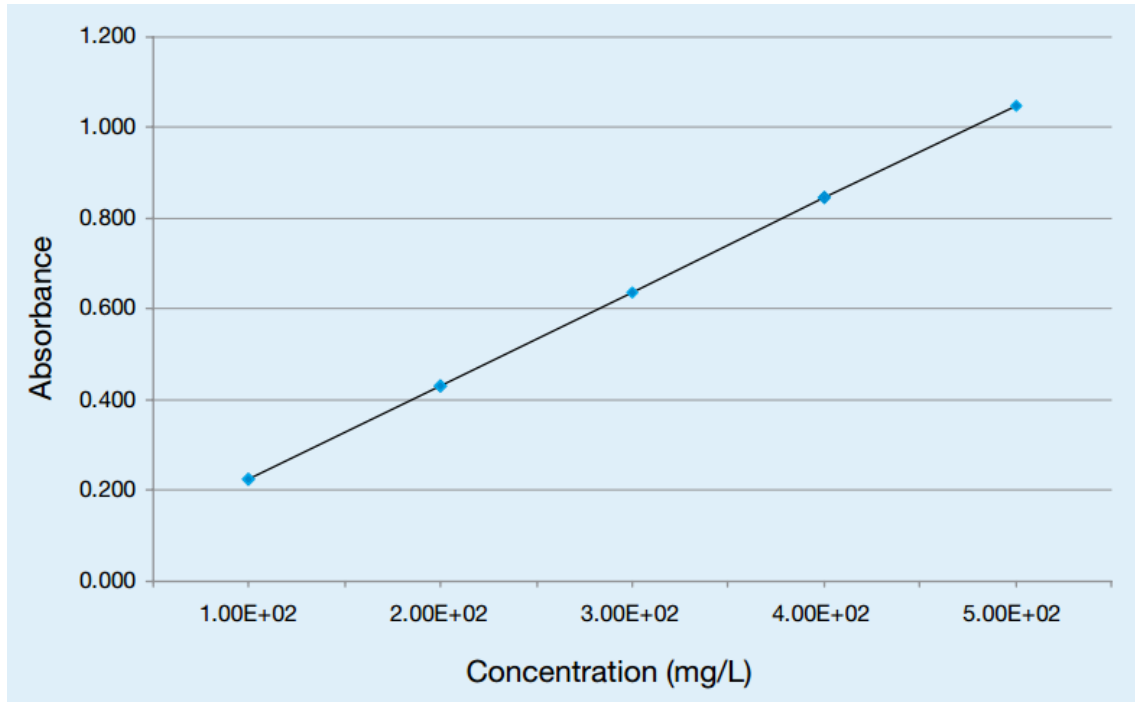


UV–VIS calibration graphs of ASP and SAL mixture

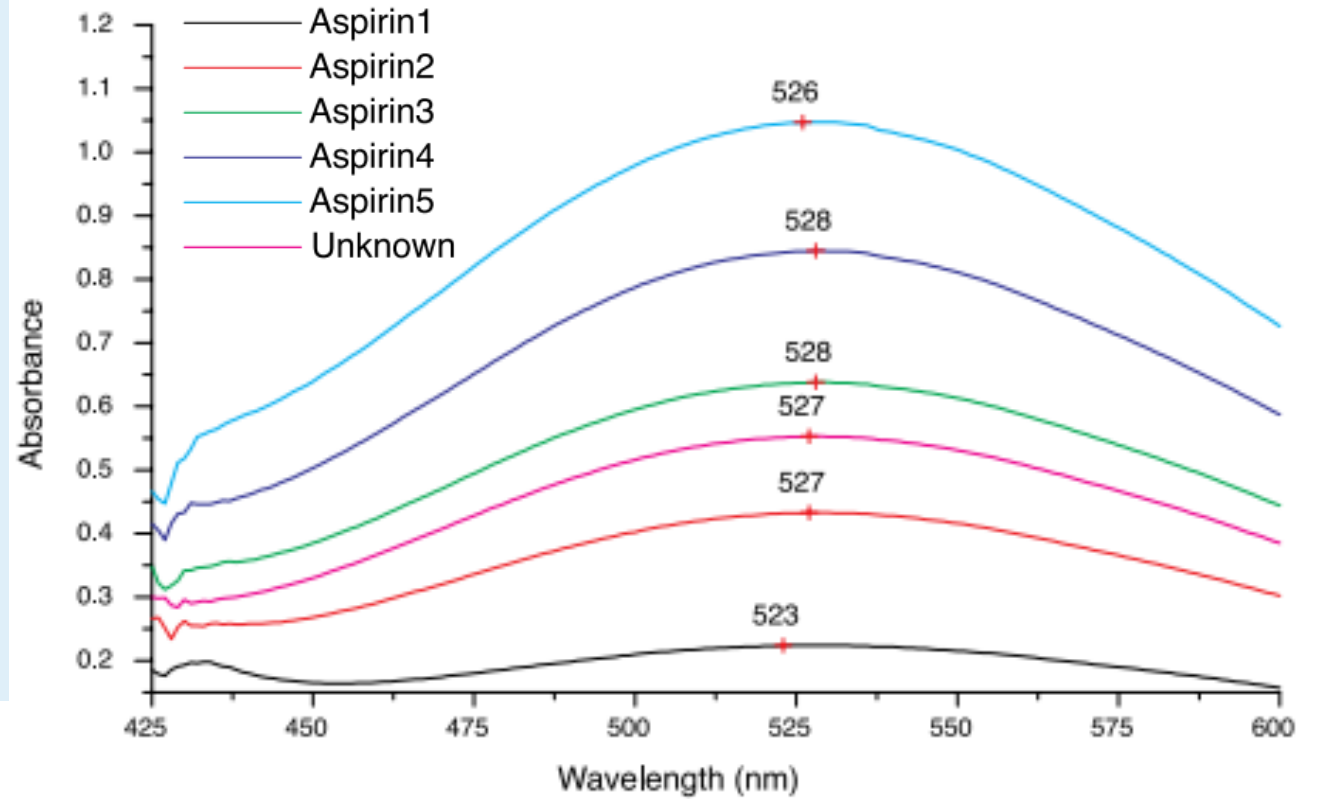
DRUG OVERDOSE STUDY

- Drug **overdose** can lead to **death**. Overdose of drug in person can be studied by UV-Vis spectroscopy.
- For example overdose of aspirin can be studied by using UV-Vis spectrum of blood plasma of a person with overdose of aspirin.
- Therapeutic levels taken after a heart attack are typically 150 – 300 mg/L and for post by-pass operations 75 mg/L. The levels of salicylate present in blood plasma can be analysed using UV-visible spectroscopy to indicate if the subject has taken a therapeutic dose or an overdose.

Therapeutic	< 300 mg/L
Moderate Overdose	500-700 mg/L
Severe Overdose	> 750 mg/L



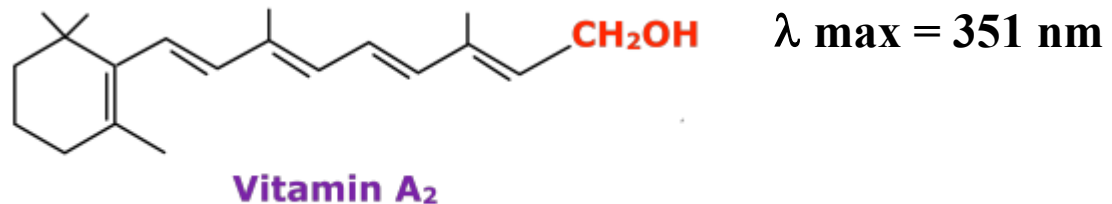
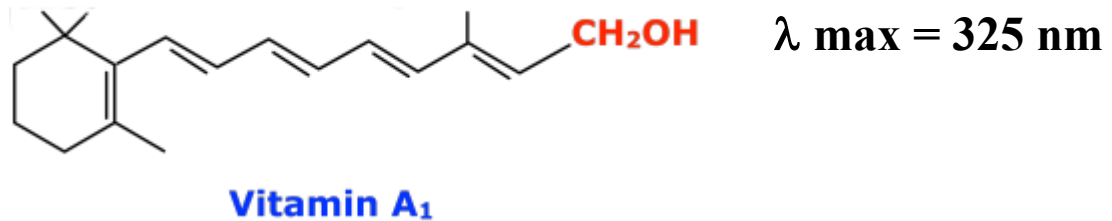
Beer-lambert calibration Plot – Salicylate in Blood Plasma



Model spectra

Quantitative analysis

Determination of concentration of the compound



The percentage of Vitamin A1 and Vitamin A2 in natural fats or oils can be estimated by the measurement of the intensities of peaks at 325 nm and at 351 nm, respectively.

A similar procedure may be used for the estimation of ergosterol in fats, anthracene in benzene, carbon disulphide in carbon tetrachloride, chlorophyll in plant material etc.

Table 1

FD&C Dye	Molar Mass (g·mol⁻¹)	ε (L·cm⁻¹·mol⁻¹)
Red 3 or Erythrosine (cherry red)	898	31,000
Red 40 or Allura Red AC (orange-red)	496	25,900
Yellow 5 or Tartrazine (lemon-yellow)	534	27,300
Yellow 6 Sunset Yellow (orange)	452	25,900
Green 3 Fast Green FCF (sea green)	809	43,000
Blue 1 Brilliant Blue FCF (bright blue)	793	130,000
Blue 2 Indigotine (royal blue; Indigo Carmine)	466	111,000

Data Table 2

Color of Solution	Dye(s) contained in solution	Pure substance or mixture?	Conc. (mol/L)
Red			
Yellow			
Green			
Blue			

Part 2. Create a Beer's Law plot for Blue No. 1 dye

Data Table 3

Solution	Dilution Ratio (mL stock/ mL water)	Molar Conc. (μM)	Measured Absorbance
1 (stock solution)	10 mL/0 mL		
2	8 mL/2 mL		
3	6 mL/4 mL		
4	4 mL/6 mL		
5	2 mL/8 mL		

Plot of Absorbance vs. Concentration for Blue No. 1 dye (Beer's Law plot)

Staple your printed graphs to this report sheet and record the required data and answers in the spaces below:

1. **Record the slope of the best-fit line:** _____

2. Write the full equation ($y = mx + b$ format) for the best fit line on the graph you just created using the slope.
3. The slope of the line is derived from the molar absorptivity (ϵ) of the dye and the path length (b) of the sample in the spectrophotometer. What is the path length of your cuvette?

Record the pathlength of your cuvette: _____ cm

4. Use the slope of the line to determine the molar absorptivity (ϵ) of Blue No. 1 dye. Use the equation for Beer's Law to derive and include the units. Note that absorbance has no units. Show your calculation here:

Part 3. What's in that drink?

Absorbance of the blue drink: _____ at _____ nm

1. **Determine the molar concentration of the Blue Dye in the drink. Show all work.**
2. **Determine the mass of Blue #1 Dye found in 591 mL of the drink. Show all work.**